Lecture 24: The thinking revolution

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will be able to...

1. Explain **test-time compute scaling** and why it represents a new paradigm for AI
2. Describe how **reasoning models** (o1, o3, DeepSeek-R1) learn to "think" via reinforcement learning
3. Compare the **frontier model landscape** as of early 2026: Claude, GPT, Gemini, open-weight models
4. Analyze **benchmark saturation** and what it tells us about progress toward general intelligence
5. Evaluate whether longer "thinking" constitutes genuine reasoning or sophisticated pattern completion

Welcome back

The last content week

This is our final week of new material. Three lectures remain:

- **Today**: The thinking revolution — reasoning models and the frontier landscape
- **Wednesday**: Agents, tools, and the agentic era
- **Friday**: The reckoning — society, safety, and what comes next

Final project reminder

Final Project presentations are **March 9** (next Monday). Submit all materials before class.

The biggest shift since transformers

A new scaling axis

Every model we've studied — from word embeddings to GPT — improved primarily by making **training** bigger: more data, more parameters, more compute during training. In late 2024, a new paradigm emerged: **test-time compute scaling** — spending more compute *at inference* to improve results.

The key insight

A smaller model that "thinks longer" can outperform a larger model that answers immediately. This decouples model quality from model size in a way that changes everything about how we build and deploy AI.

Two scaling axes

Training compute vs. inference compute

	Training compute	Inference compute
When	Before deployment (once)	At query time (every call)
Cost structure	Fixed, upfront, massive	Per-query, variable
What improves	Base knowledge, capabilities	Reasoning depth on hard problems
Scaling law	Kaplan et al. (2020) — Lecture 16	Snell et al. (2024) — **new**
Analogy	Years of education	Minutes of careful thought

The analogy

Training compute is like years of schooling — it determines what you *know*. Inference compute is like time spent thinking about a specific problem — it determines how *carefully* you reason. Both matter, independently.

Test-time compute scaling

Snell et al. (2024): 'Scaling LLM Test-Time Compute'

Snell et al. (2024) showed that **additional compute at inference** improves performance predictably and substantially — and on some problems, a small model thinking longer outperforms a large model thinking quickly.

Two mechanisms:

1. **Process-reward models**: Dense, step-level verifiers that score each reasoning step
2. **Adaptive self-revision**: The model critiques and refines its own output iteratively

The empirical result

Compute-optimal test-time scaling improves efficiency **2–4×** over naive approaches. On certain problems, spending additional FLOPs at inference exceeds the benefit of spending the same FLOPs on pretraining.

From chain-of-thought to reasoning models

The evolution

Year	Technique	Key idea
2022	Chain-of-thought prompting	Show the model examples with reasoning steps
2023	Tree-of-thought, self-consistency	Generate multiple reasoning paths, pick the best
2024	**Reasoning models** (o1)	Train the model via RL to generate its own reasoning
2025	Adaptive thinking (Claude 4.x)	Model decides *when and how much* to think

The conceptual leap

Chain-of-thought prompting (Lecture 16) showed that intermediate reasoning helps. Reasoning models take this further: instead of *prompting* the model to reason, you **train it via reinforcement learning** on verifiable rewards (correct math, passing code tests). The model learns to generate its own internal reasoning traces — and these traces can be far more effective than human-written examples.

OpenAI o1: the first reasoning model

OpenAI (September 2024)

o1 was trained via large-scale RL where the reward signal is *outcome correctness* — verified math solutions, code that passes unit tests. The model learns to generate an internal chain-of-thought that maximizes this reward.

How o1 works

1. User submits a question
2. Model generates a long internal **thinking trace** (hidden from user, billed as tokens)
3. Thinking trace includes reasoning steps, self-correction, backtracking
4. Model produces a final answer based on its thinking
5. **No tree search or MCTS at inference** — all "search" happens during RL training

Key limitation

OpenAI hides o1's thinking tokens and forbids prompting the model to reveal them. Whether the model performs genuine systematic reasoning or fluent post-hoc rationalization remains an open research question.

o3 and o4-mini: reasoning scales up

OpenAI (April 2025)

Model	AIME 2025	GPQA Diamond	Codeforces Elo	Key advance
GPT-4o (baseline)	~13%	~53%	~1200	No reasoning
o1 (Sept 2024)	~74%	~78%	~1900	First reasoning model
o3 (Apr 2025)	88.9%	87.7%	2727	20% fewer errors than o1
o4-mini (Apr 2025)	**92.7%**	~87%	—	Multimodal + tool use in reasoning loop

What changed

o3 makes **20% fewer major errors** than o1. o4-mini — the *small* reasoning model — scores **92.7% on AIME 2025** (math olympiad problems), surpassing o1's 74%. And o4-mini adds native multimodal input and tool use *within* the reasoning loop — it can search the web and run code as part of its thinking process.

DeepSeek-R1: open-source reasoning

DeepSeek-AI (January 2025) — MIT License

DeepSeek-R1 proved that frontier reasoning capabilities can be achieved with **pure reinforcement learning** on an open-weight model — no proprietary data pipelines, no human preference labels.

The R1-Zero experiment

DeepSeek-R1-Zero was trained with RL only (zero supervised fine-tuning):

- Algorithm: **GRPO** (Group Relative Policy Optimization) — generates multiple answers, ranks by correctness, updates toward better ones
- Reward: Only whether the final answer is correct — no step-level supervision
- Result: Self-verification, reflection, and long chain-of-thought **emerged spontaneously**
- AIME 2024: jumped from 15.6% → 71.0% (pass@1); 86.7% with majority voting

Why this matters

R1 matched OpenAI o1 as a fully open model. Its API costs ~1/30th of o1. Six distilled versions (1.5B–70B) outperform GPT-4 on math/coding. This democratized reasoning.

Claude's extended thinking

Anthropic (February 2025 — present)

Anthropic implemented reasoning as a **toggle within a single model** — same weights, different inference behavior. Unlike OpenAI, Claude's thinking tokens are **visible** to developers.

API usage

```
response = client.messages.create(
    model="claude-opus-4-6",
    thinking={"type": "enabled", "budget_tokens": 10000},
    messages=[{"role": "user", "content": "Prove that √2 is irrational."}]
)
# Response includes a visible "thinking" block + final answer
```

Key differences from OpenAI

Feature	OpenAI o-series	Claude extended thinking
Thinking visibility	Hidden	**Visible**
Control mechanism	Reasoning effort (low/med/high)	Budget tokens (1K–128K) or adaptive

The s1 experiment: reasoning is surprisingly simple

Muennighoff et al. (2025)

s1 showed that test-time scaling can be achieved with remarkably little effort:

1. Fine-tune Qwen2.5-32B on just **1,000 curated examples** (the "s1K" dataset)
2. At inference, apply **budget forcing**: append the word "Wait" to force continued reasoning
3. Result: Exceeds o1-preview on AIME 2024 by up to **27%**

The implication

You don't need massive RL infrastructure to get reasoning capabilities. A small amount of high-quality reasoning data + a simple inference trick can unlock substantial test-time scaling. This suggests reasoning is latent in large pretrained models — it just needs to be activated.

The frontier landscape: early 2026

Where we stand

Model	Developer	Release	Total params	Key capability
Claude Opus 4.6	Anthropic	Feb 2026	—	1M context, adaptive thinking
GPT-5	OpenAI	Aug 2025	—	Native multimodal, 94.6% AIME
Gemini 2.5 Pro	Google	Mar 2025	—	1M context, built-in thinking
DeepSeek-R1	DeepSeek	Jan 2025	671B MoE	Open-weight reasoning, $5.5M training
Llama 4 Maverick	Meta	Apr 2025	400B MoE	Open-weight, 1M context, multimodal
Qwen 3	Alibaba	Apr 2025	235B	89.7% AIME, open-weight
Mistral Large 3	Mistral AI	Dec 2025	675B MoE	Open-weight, 256K context, 40+

The Mixture of Experts revolution

MoE is now the default for frontier models

Nearly every frontier model released in 2025 uses **Mixture of Experts** (MoE) — sparse activation where only a fraction of parameters are used per token:

Model	Total params	Active params	Experts	Training cost
DeepSeek-V3	671B	37B	256 + shared	**$5.6M**
Llama 4 Maverick	400B	17B	128 + 1 shared	—
Llama 4 Scout	109B	17B	16	—
Qwen 3	235B	22B	128 (top-8)	—
Mistral Large 3	675B	41B	—	—

The DeepSeek effect

DeepSeek-V3 trained a frontier model for ~$5.6M — roughly **1/20th** of GPT-4's estimated cost. This shattered the assumption that only billion-dollar labs can compete. The key innovations: Multi-head Latent Attention (MLA), auxiliary-loss-free load balancing, and multi-token prediction.

Benchmark saturation

We're running out of tests

Most standard benchmarks are now **saturated** — frontier models score at or above human expert level:

Benchmark	Status	Best score	Human baseline
HumanEval (coding)	Saturated	99.0%	~95%
MMLU (knowledge)	Saturated	~92%	~89% expert
GPQA Diamond (PhD-level science)	Near-saturated	94.3%	~65% expert
MATH-500	Near-saturated	98.0%	—
SWE-bench Verified (real GitHub issues)	Active	80.9%	—
ARC-AGI-2 (novel reasoning)	**Not saturated**	**4.4%**	**60%**
Humanity's Last Exam	**Not saturated**	**48.1%**	**~90%**

Humanity's Last Exam

2,500 expert-level questions across 100+ academic subjects

Humanity's Last Exam (HLE) was designed to be the hardest public benchmark — questions submitted by domain experts that require deep specialist knowledge.

Progress in one year

System	HLE score	Date
Early 2025 frontier models	~1–5%	Jan 2025
OpenAI Deep Research	26.0%	Feb 2025
Claude Opus 4.6 (max thinking)	36.7%	Feb 2026
Gemini 3.1 Pro Preview	44.7%	Feb 2026
SOTA (Zoom AI)	**48.1%**	Late 2025
Human graduate students	**~90%**	—

What this tells us

Models went from ~3% to ~48% in one year — extraordinary progress. But the gap to human experts (~90%) remains large. HLE and ARC-AGI-2 suggest that while models excel at knowledge retrieval and pattern-matching, **novel reasoning and deep domain expertise** remain hard.

ARC-AGI: the reasoning gap

Two versions, two very different stories

ARC-AGI-1 (Chollet, 2019): Visual pattern puzzles requiring novel reasoning.

- o3 scored **87.5%** at high compute ($17–20/puzzle) — near human level (95%)
- Declared "largely solved" → prize retired

ARC-AGI-2 (2025): Harder puzzles, same format.

- o3 scored **2.9%** — humans score **60%**
- The **20× gap** between AI and humans on novel tasks remains enormous

The lesson

Standard benchmarks can be saturated by scale and pattern matching. But **genuinely novel reasoning** — the kind that requires understanding abstract principles and applying them to situations never seen before — remains the frontier. ARC-AGI-2 is arguably the best current test of this capability.

Native multimodal models

Models that see, hear, and generate across modalities

A parallel revolution: frontier models are no longer text-only. They **natively** process and generate across modalities:

Capability	GPT-5	Gemini 2.5+	Claude 4.6	Llama 4
Text input/output	Yes	Yes	Yes	Yes
Image understanding	Yes	Yes	Yes	Yes
Audio input	Yes	Yes	No	No
Video understanding	Frames	**Native**	No	Yes
Image generation	**Native**	**Native**	No	No
Long context	128K	**1M**	**1M**	1M

Native image generation (March 2025)

Both OpenAI and Google shipped **native image generation** inside their main LLMs — not calling a separate diffusion model, but generating images autoregressively as tokens. GPT-4o generates images with far superior text rendering vs. DALL-E 3. Gemini does the same with SynthID watermarking built in.

Small models, big capabilities

Not everyone needs 671B parameters

Model	Parameters	Key strength
Phi-4-mini (Microsoft)	3.8B	Matches 7–9B class on reasoning
Gemma 3n (Google)	2B effective	Multimodal (text+image+audio+video), on-device
Qwen 3-0.6B (Alibaba)	600M	Hybrid reasoning modes, tool use
Llama 3.2 1B (Meta)	1B	128K context, fits in 2–3 GB RAM

The inference-optimal paradigm

These models are **massively overtrained** relative to Chinchilla-optimal scaling (Lecture 16): 700+ tokens/parameter vs. the "optimal" 20. The logic: train once, deploy millions of times. Smaller models are cheaper to run — and with quantization, a 3B model can run on a **phone**.

On-device AI

LLMs in your pocket

Flagship smartphones in 2025 can run 4B+ parameter models at conversational speeds:

Platform	NPU capability	What runs locally
Apple A18 Pro	35+ TOPS	~3B model; sensitive tasks stay on-device
Qualcomm Snapdragon 8 Elite	70+ TOPS	Llama 3.2 1B/3B natively
MediaTek Dimensity 9500	3nm NPU	Gemma 3n via LiteRT

Apple's privacy architecture

Apple Intelligence uses a tiered approach:

1. **On-device**: ~3B model handles simple tasks privately
2. **Private Cloud Compute**: Complex tasks processed on Apple silicon servers — no data stored
3. **External**: ChatGPT integration for out-of-scope tasks (with user permission)

This is a principled answer to the privacy vs. capability tradeoff.

Voice and real-time AI

End-to-end audio models

GPT-4o Advanced Voice Mode processes audio natively — no speech-to-text → LLM → text-to-speech pipeline. One unified model:

- Detects sarcasm, urgency, hesitation from **acoustic signals**
- Handles natural **interruptions** (you can cut it off mid-sentence)
- Sub-second latency via WebRTC
- Modulates its own voice emotionally based on context

Gemini Live adds deep Google ecosystem grounding — it can check your Gmail, Calendar, and Drive while talking to you.

Questions to consider

When an AI can hear your tone of voice, see your face, and respond in real-time with emotional awareness — is this meaningfully different from human conversation? What does this mean for the ELIZA effect (Lecture 2)?

Discussion: what does it mean to "think"?

The deepest question in this course

1. **Thinking or performing?** When o3 generates a 10,000-token reasoning trace to solve a math problem, is it *thinking* — or producing a sophisticated pattern that *looks like* thinking? How would you test the difference? (Revisit Lecture 1: is ChatGPT conscious?)
2. **The DeepSeek-R1 emergence:** R1-Zero developed self-verification and backtracking *without being taught these behaviors*. The model was rewarded only for correct answers — yet it learned to doubt itself, re-examine its work, and try alternative approaches. Is this "just optimization" — or is something deeper happening?
3. **Diminishing returns?** AIME scores went from 13% → 93% in 18 months. But ARC-AGI-2 scores went from 0% to 4%. Are we approaching a wall — or just need a different kind of reasoning

Further reading

Further reading

Snell et al. (2024, *arXiv*) "Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters" — The theoretical foundation for inference-time scaling.

DeepSeek-AI (2025, *arXiv*) "DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning" — Open-weight reasoning model matching o1.

Muennighoff et al. (2025, *arXiv*) "s1: Simple Test-Time Scaling" — 1,000 examples + "Wait" trick beats o1-preview.

Wei et al. (2022, *NeurIPS*) "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models" — Where it all started.

OpenAI (2025) "Introducing o3 and o4-mini" — Latest reasoning models with multimodal capabilities.

Questions?

[Email] [Discord] [Office hours]

Up next...

Agents, tools, and the agentic era: when LLMs start acting in the world